AN UNSUPERVISED DEEP LEARNING METHOD FOR THE SUPER-RESOLUTION OF RADAR SOUNDER DATA

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ABSTRACT

Radar sounders (RSs) are widely used to image profiles (radargrams) of the subsurface of planetary bodies and the Earth. However, despite the huge scientific return from radargram analyses, their horizontal and vertical resolutions are limited by technical factors. Even if methods exist for improving the resolution, these are still limited by technical factors and introduce artifacts. This paper proposes an unsupervised deep-learning method that synthesizes accurate super-resolved radargrams overcoming these limitations. The method adopts the Cycle-Consistent Adversarial Network (CyleGAN) that learns the mapping function between the low- and high-resolution data distributions. The network is adapted to match the low- and high-resolution radargram characteristics, including the differences in dimensions and radiometric properties. The proposed method was successfully validated on airborne data at higher resolution and simulated data with lower resolution.

Index Terms— super-resolution, deep learning, generative models, adversarial neural networks, radar sounder.

1. INTRODUCTION

Radar sounders (RSs) image the subsurface by transmitting electromagnetic (EM) waves at the nadir with a frequency in the range of high or very high frequencies (HF-VHF) and relatively wide bandwidth. Such waves propagate into the subsurface and interact with the buried dielectric interfaces, generating backscattered echoes. The echoes are collected by the antenna and represented in radargrams. Although radargrams are important for better understanding subsurface geological structures and processes, there are technical limitations that reduce their quality in terms of azimuth (horizontal) and range (vertical) resolution [1]. The azimuth resolution can be approximated as the antenna footprint size in the along-track direction [1]. RS antenna normally has a broad radiation pattern, a very large ground footprint, and thus a low azimuth resolution. The azimuth resolution can be improved by exploiting the Doppler effect and the synthetic aperture processing that, however, is limited by the synthetic antenna length [1]. Assuming signals transmitted with linear frequency modulation, the range resolution is proportional to the bandwidth and the frequency. The latter is expected to be low enough for a deep penetration, which conflicts with the need for a wide bandwidth for a finer range resolution [1]. BandWidth Extrapolation (BWE) methods can mitigate this issue and are based on the interpolation and extrapolation of the signal spectrum with a linear prediction model [2]. Although BWE methods showed promising results on planetary radargrams, they may introduce artifacts (e.g., line splitting and undesired reflections).

Recently, several methods (called super-resolution methods) have been proposed in the remote sensing literature to enhance data resolution considering prior information to reduce artifacts. Some methods exploit the texture and the spatial information in the data to train generative models to accurately super-resolve low-resolution data. During the training, deep networks encode the spatial and radiometric properties of the low- and high-resolution images. Hence, the network accurately reconstructs higher resolution images. However, although these methods showed promising results on several remote sensing images, they were never used on radargrams.

This paper proposes an unsupervised method based on generative deep-learning models for an accurate super-resolution of radar sounder data. The generative model learns the mapping function between the distributions of the low- and high-resolution radargrams that can be used to solve the ill-posed problem of super-resolving radargrams not seen before. Here, we exploit the Cycle-Consistent Generative Adversarial Network (CycleGAN) as it processes unpaired images and handles complex texture [3]. We adopted the architecture in [3] and adapted it to match the properties of radargrams at different resolutions, including the varying size. The proposed method was successfully validated on airborne RS data and simulated RS data at a lower resolution in icy areas. The network trained on these icy radargrams can be used to super-resolve other low-resolution radargrams, including planetary data, acquired in an icy environment.

2. PROPOSED SUPER-RESOLUTION METHOD

Let us consider two radargrams $\mathbf{R_L}$ and $\mathbf{R_H}$ acquired by two RSs with different system parameters so that the range and

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azimuth resolution of $\mathbf{R_L}$ are lower than those of $\mathbf{R_H}$. Each radargram can be considered as a 2D matrix of N_T traces and N_S samples defined as $\mathbf{R} = \{R(a,r)|a \in [1,...,N_T], r \in [1,...,N_S]\}$, where R is the power of the reflections in the radargrams, and (a,r) the azimuth and the range coordinates, respectively and corresponds to a trace of the radargram. Adjacent pixels capture the same geologic targets in a given area of the subsurface. Given the different RS parameters, the number of traces N_T^L and N_T^H and samples N_S^L and N_S^H of two radargrams covering the same along-track length and range penetration are different, i.e., $N_T^L < N_T^H$.

The proposed approach exploits generative deep models that can synthesize new data by learning the mapping function between the distributions of the input low-resolution $\mathcal{X}$ domain and output high-resolution $\mathcal{Y}$ domain. The mapping function can be used to accurately super-resolve radargrams not previously seen. The proposed method is completely unsupervised and does not require pairs of high and low-resolution patches sharing the same ground track. We adopted the cycle-consistent generative adversarial network (CycleGAN) since it processes unpaired images and handles complex textures. We adapted the CycleGAN in [3] to match the radar data properties at different resolutions, including the size. The method consists of three steps– we first modify the CycleGAN to match the properties of the data at different resolutions. Secondly, we train the modified CycleGAN to learn i) the spatial and radiometric properties of $\mathcal{X}$ and $\mathcal{Y}$, and ii) the mapping function between $\mathcal{X}$ and $\mathcal{Y}$. Then, the mapping function can be used for accurately super-resolving unseen radargrams.

2.1. Adaptation of the network architecture

As CycleGAN architecture, we considered that in [4] as it handles the radar data, including the signal and noise properties and the one channel dimension. A CycleGAN consists of two GAN networks, each made of a generator and a discriminator that can have different structures. We adopt the discriminator architecture presented in [4] that consists of six convolutional layers. The generator architecture in [4] is modified to match the different sizes of the low- and high-resolution data. We modify the generator architecture by adding convolutional layers. This results in two asymmetric networks that handle data with different resolutions and dimensions.

2.2. CycleGAN training formulation

The generator and discriminator are simultaneously trained with an adversarial loss $\mathcal{L}_{GAN}$. The discriminator learns the differences between real and fake input data, while the generator aims at synthesizing realistic images that can fool the discriminator. As an extension of GAN, CycleGAN learns features from two different domains $\mathcal{X}$ and $\mathcal{Y}$ to perform domain translation. Thus, the system learns two mapping functions $G : \mathcal{X} \rightarrow \mathcal{Y}$ and $F : \mathcal{Y} \rightarrow \mathcal{X}$ so that the translation

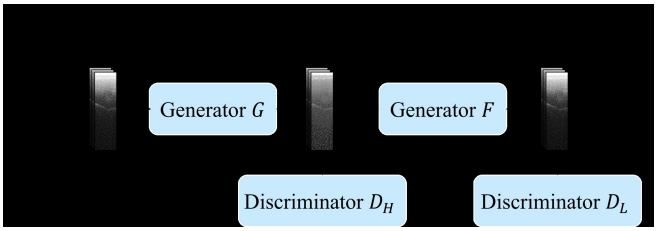


Fig. 1: Diagram of the CycleGAN training.

is bidirectional. After translating an image from $\mathcal{X}$ to $\mathcal{Y}$, the additional mapping function F can reconstruct the image in $\mathcal{X}$ (see Fig. 1) with the cycle consistency loss $\mathcal{L}_{cyc}$. Firstly, adversarial losses are applied to assess that each generator associated with the related discriminator D_L or D_H learns mapping functions G and F. The adversarial loss for the mapping function G is defined as

$$\mathcal{L}_{GAN}(G, D_H, \mathcal{X}, \mathcal{Y}) = E_{\mathbf{R_H} \sim pdata}(\mathbf{R_H})[\log D_H(\mathbf{R_H})] + E_{\mathbf{R_L} \sim pdata}(\mathbf{R_L})[\log(1 - D_H(G(\mathbf{R_L})))] \quad (1)$$

where i) the generator G synthesizes the image $G(\mathbf{R_L})$ that has the same distribution of the images in $\mathcal{Y}$, and ii) the discriminator distinguishes between the translated samples $G(\mathbf{R_L})$ and the real sample $\mathbf{R_H}$. Hence, the generator minimizes the loss function while the discriminator maximizes it. Similarly, we can define the adversarial loss function $\mathcal{L}_{GAN}(F, D_L, \mathcal{X}, \mathcal{Y})$ to learn the generator mapping function F and the discriminator D_L. Secondly, to make the mapping functions F and G the reverses of each other, the cycle consistency loss $\mathcal{L}_{cyc}(G, F)$ is introduced to force $F(G(\mathbf{R_L})) \approx \mathbf{R_L}$ and $G(F(\mathbf{R_H})) \approx \mathbf{R_H}$. $\mathcal{L}_{cyc}(G, F)$ reduces the space of possible mapping functions by enforcing forward and backward consistency as follows:

$$\mathcal{L}_{cyc}(G, F) = E_{\mathbf{R_L} \sim pdata}(\mathbf{R_L})[\|F(G(\mathbf{R_L})) - \mathbf{R_L}\|] + E_{\mathbf{R_H} \sim pdata}(\mathbf{R_H})[\|G(F(\mathbf{R_H})) - \mathbf{R_H}\|] \quad (2)$$

Hence, the full objective loss of CycleGAN is defined as:

$$\mathcal{L}(G, F, D_L, D_H) = \mathcal{L}_{GAN}(G, D_H, \mathcal{X}, \mathcal{Y}) + \mathcal{L}_{GAN}(G, D_L, \mathcal{X}, \mathcal{Y}) + \lambda \mathcal{L}_{cyc}(G, F) \quad (3)$$

where λ controls the relevance of $\mathcal{L}_{cyc}(G, F)$. Once the CycleGAN is trained, the mapping function G can be used to super-resolve previously unseen low-resolution radargrams. If needed, the mapping function F can be used to reduce the resolution of unseen high-resolution radargrams.

3. DATASETS AND EXPERIMENTAL RESULTS

To assess the performance of the proposed method, we applied it to a dataset of two sets of radargrams imaging icy

Table 1: Parameters of the EORS [6] and ARS [5] instruments and the related radargrams.

Parameters	EORS [6]	MCoRDS [5]
Range resolution in ice	10 m	4.3 m
Focused azimuth resolution	1.3 km	25 m
Central frequency [MHZ]	45	195
Bandwidth [MHZ]	10	30
Peak radiated power	800 W	1 kW
Orbit height	500 km	500 m

areas. MCoRDS-3 acquires the first set [5] over Greenland in 75 campaigns between 2013 and 2017. The second dataset is generated by simulating Earth-orbiting radar sounder (EORS) data from the MCoRDS data with the analog-based simulator in [6]. Table 1 shows the instrument parameters of EROS and MCoRDS and the related resolutions. MCoRDS data have higher resolutions than simulated EORS data. MCoRDS and EORS radargrams have been preprocessed with range compression, and azimuth focusing via synthetic aperture radar techniques, and the clutter is partially compensated. Moreover, we mitigated the artifacts due to airplane fluctuations. We also normalized the power with respect to the surface for each column, and we applied the log-transformation to enhance the spatial information. Finally, we divided the MCoRDS and EORS log-power radargrams into patches to have two sets of about 4000 patches. MCoRDS patches have 256 traces and 1280 samples, while the EROS patches have 64 traces and 1920 samples. The along-track dimension is chosen so the ground track lengths of patches are the same (i.e., the patches cover the same length). Finally, all the patches are standardized by removing the mean and dividing by the standard deviation. To increase the network invariance and robustness given the relatively small training set, we perform data augmentation to generate larger datasets [4]. We apply standard data augmentation techniques (i.e., random horizontal flips, rotation, elastic deformation, and grid distortion). The transformations are designed to preserve the subsurface target geometrical properties and to have realistic augmented patches, like in [7].

To assess the robustness of the proposed technique, we performed two experiments by varying i) the network architecture used as a generator and ii) the percentage of samples in the training set. For the first experiment, we considered two types of networks, i.e., a residual network (ResNet) modified from [3] and the U-Net presented in [4]. For the residual network, we performed two experiments by fixing i) all the kernel dimensions to 3×3, and ii) using different kernel sizes in the convolutional layers. In addition, we considered a different percentage of EORS-MCoRDS training pairs for the second experiment, i.e., 80%, 70%, and 50%. A random 10% of the training set is used as a validation set. The training, validation, and test sets are independent

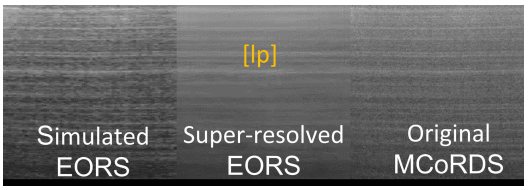


Fig. 2: Simulated and super-resolved EORS patches compared with the original MCoRDS patch with a ResNet generator and 80% of the samples for the training.

for all experiments. At each epoch of the training, a small batch of patches is given as input to the network to accelerate the training phase. For this reason and memory constraints, the mini-batch size is set equal to 12. For the optimization, we used Adam as it shows good results in the literature [7]. The learning rate is set to $\ell_r = 5 \cdot 10^{-7}$ for the ResNet and $\ell_r = 2 \cdot 10^{-5}$ for the U-Net after running several experiments varying its values from 0.1 to 10^{-9}. Finally, the training stop condition is based on the maximum epochs number set to 100. To quantitatively assess the effectiveness of the proposed method, we compared the real MCoRDS data and the super-resolved patches in terms of signal-to-noise ratio (SNR), peak signal-to-noise ratio (PSNR), mean square error (MSE), signal-to-reconstruction error (SRE) in dB, structural similarity index (SSIM), and normalized cross-correlation coefficient (NCC). MSE, SSIM, and NCC are expected to be in the range of $[0, 1]$ – lower the MSE, the better is the reconstruction. On the contrary, higher values of SSIM and NCC indicate a better super-resolved patch. Finally, PSNR, SNR, and SRE indices indicate a good reconstruction when their value for the super-resolved and the original patch are close.

The proposed method generally shows promising results in super-resolving the EORS patches. The method fairly synthesizes the geometric and radiometric properties of the bedrock and the behavior of the ice stratigraphy. Fig. 2, 3, and 4 show examples of patches pointing out some critical issues with different experimental setups. A low pass effect on the reflections is visible in Fig. 2 and 4 (label *lp*), including the stratigraphy and the bedrock, due to a poor modeling of the signal statistics. Fig. 3 and 4 show artifacts in the super-resolved images (labeled with *a1* and *a2*) that depend on the EORS data quality and the fact that in some EORS radargrams the information is not enough for the unmixing needed to reconstruct the ice stratigraphy. The U-Net introduces larger artifacts than the ResNet due to skip-connections adding basic spatial features to the latest network layers. Table 2 shows the performance of the different experimental setups. The SSIM and NCC indices suggest a good structural correlation between the EORS simulated and super-resolved images. The experiments with the residual network as a generator show better results than those with the U-Net for most metrics. The ResNet-based results show that a fixed kernel size is limiting, and the super-resolved patches improve by

Table 2: Performance of the proposed method by varying the network architecture for the generator.

Index	ResNET Dataset 80%	ResNET Dataset 70%	ResNET Dataset 50%	ResNET Kernel 3×3	U-NET	Reference Value
PSNR	38.2390	37.5830	36.4022	31.7389	35.2837	38.2509
SNR	13.9729	13.2909	12.1811	12.9945	11.0229	14.3454
SRE	8.6252	8.1225	7.2396	8.7664	7.6368	8.6183
MSE	0.000284	0.000290	0.000314	0.00090	0.000427	0
SSIM	0.8614	0.8624	0.8557	0.69431	0.7989	1
NCC	0.9765	0.9762	0.9200	0.9579	0.9471	1

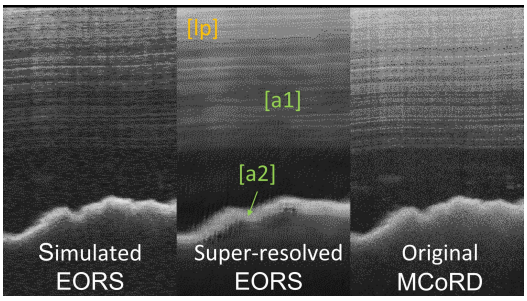


Fig. 3: Simulated and super-resolved EORS patches compared with the original MCoRDS patch considering a ResNet generator with fixed 3×3 kernel filters.

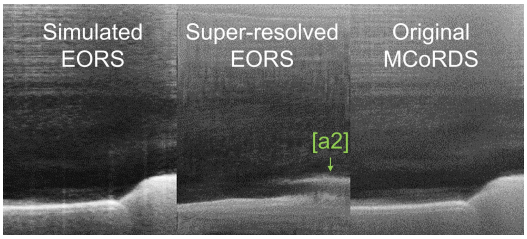


Fig. 4: Simulated and supe-resolved EORS patches compared with the original MCoRDS patch using a U-Net [4] generator.

setting different kernel sizes within the same network. This is because filters with different receptive fields can extract features with different-sized details.

4. CONCLUSIONS

The paper proposed a deep-learning-based method for generating super-resolved radargrams in an unsupervised way. The method solves a strongly ill-posed problem by exploiting generative models trained to learn the geometric and radiometric properties of the radargrams and the mapping function between the low- and high-resolution domains. Such mapping function can be used to super-resolve previously unseen low-resolution radargrams. The method is applicable when the low and high resolution data are imaging similar geological targets. Considering this constraint, the method can be used for super-resolving any radar sounder data (e.g., future STRATUS [8] couplet with MCoRDS). The method showed promising results in validating low-resolution simulated EORS and high-resolution MCoRDS data. However, it must be further developed to constrain better the reconstruction avoiding artifacts and better model the signal statistics. Note that given that the super-resolution problem is strongly ill-posed, the architecture choice, the training set size, and the selection of the network parameters (e.g., kernel dimension and receptive field) are crucial because they define the constraints in the super-resolution process. As future work, we plan to investigate how to mitigate the low-pass effect and generate super-resolved patches with finer details. Moreover, we plan to use a larger multi-sensor training dataset that may help to reduce the size and number of artifacts.

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